WORKED SOLUTION

STEP I 2010, Question 12

Probability and Statistics • Expectation via Tail Sums

STEP I 2010, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s1-q12

The key idea

The tail-sum formula comes from writing $n\,\mathrm{P}(X = n)$ as $\mathrm{P}(X = n)$ added up n times, and then collecting the terms in a different order. For the penguins, the event $X \geq n$ is much easier to describe than $X = n$: you still need to open more boxes after $n - 1$ of them exactly when those $n - 1$ boxes all contain the same kind of penguin. So the tail probabilities are simple, and the formula we have just proved turns $\mathrm{E}(X)$ into two geometric series.

Solution

The definition and the tail-sum formula

Since X takes only the values $1, 2, 3, \ldots$,

$$\mathrm{E}(X) = \sum_{m=1}^{\infty} m\,\mathrm{P}(X = m),$$

provided this series converges.

For each $n \geq 1$, the event $X \geq n$ is the disjoint union of the events $X = m$ for $m \geq n$, so $\mathrm{P}(X \geq n) = \sum_{m \geq n} \mathrm{P}(X = m)$. Hence

$$\begin{aligned}\sum_{n=1}^{\infty} \mathrm{P}(X \geq n) &= \sum_{n=1}^{\infty} \sum_{m=n}^{\infty} \mathrm{P}(X = m) \\ &= \sum_{m=1}^{\infty} \sum_{n=1}^{m} \mathrm{P}(X = m) \\ &= \sum_{m=1}^{\infty} m\,\mathrm{P}(X = m) = \mathrm{E}(X).\end{aligned}$$

Ask yourself Why are we allowed to swap the order of the two sums?

The double sum adds $\mathrm{P}(X = m)$ over all pairs (n, m) with $1 \leq n \leq m$. In the first line we fix n and sum over m; in the second we fix m and sum over the m values $n = 1, \ldots, m$. Every term is non-negative, and for a series of non-negative terms the total does not depend on the order of summation (both sides may be infinite, in which case both are). It helps to picture the terms laid out in a triangle with row m containing m copies of $\mathrm{P}(X = m)$: summing by rows gives $\mathrm{E}(X)$, summing by columns gives the tail sum. This proves the formula.

The tail probabilities

Assume, as is intended, that the contents of different boxes are independent. Let D_k be the event that box k contains a daddy penguin, so $\mathrm{P}(D_k) = p$ and $\mathrm{P}(D_k') = q$.

Take $n \geq 2$. We have $X \geq n$ exactly when the first $n-1$ boxes do *not* yet contain one of each kind, that is, when they are all daddies or all mummies. These two cases are disjoint, so by independence

$$\mathrm{P}(X \geq n) = p^{n-1} + q^{n-1} \qquad (n \geq 2).$$

In particular, with $n = 4$,

$$\mathrm{P}(X \geq 4) = p^3 + q^3,$$

as required. Also $X \geq 2$ always, since one box can't hold both kinds, so $\mathrm{P}(X \geq 1) = \mathrm{P}(X \geq 2) = 1$. (The formula above gives $p + q = 1$ at $n = 2$, which agrees, but it gives 2 at $n = 1$, so the $n = 1$ term must be treated separately.)

The expectation

By the tail-sum formula,

$$\begin{aligned}
\mathrm{E}(X) &= \mathrm{P}(X \geq 1) + \sum_{n=2}^{\infty} (p^{n-1} + q^{n-1}) \\
&= 1 + \sum_{k=1}^{\infty} p^k + \sum_{k=1}^{\infty} q^k \\
&= 1 + \frac{p}{1-p} + \frac{q}{1-q},
\end{aligned}$$

where both geometric series converge because $0 < p < 1$ and $0 < q < 1$ (this is where $p \neq 0$, $q \neq 0$ is used). Since $1 - p = q$ and $1 - q = p$,

$$\mathrm{E}(X) = 1 + \frac{p}{q} + \frac{q}{p} = 1 + \frac{p^2 + q^2}{pq}.$$

Now $p^2 + q^2 = (p+q)^2 - 2pq = 1 - 2pq$, so

$$\mathrm{E}(X) = 1 + \frac{1 - 2pq}{pq} = 1 + \frac{1}{pq} - 2 = \frac{1}{pq} - 1,$$

as required.

The final part

By the AM–GM inequality for the positive numbers p and q,

$$\sqrt{pq} \le \frac{p+q}{2} = \frac{1}{2}, \qquad \text{so} \qquad 0 < pq \le \frac{1}{4}.$$

Ask yourself Which way does the inequality go when we take reciprocals?

Since $pq > 0$, taking reciprocals reverses the inequality: $\dfrac{1}{pq} \ge 4$. Therefore

$$\mathrm{E}(X) = \frac{1}{pq} - 1 \ge 4 - 1 = 3,$$

with equality exactly when $p = q = \frac{1}{2}$. (Equivalently, $pq = p(1-p) = \frac{1}{4} - (p - \frac{1}{2})^2 \le \frac{1}{4}$.)

Checks

- With $p = q = \frac{1}{2}$: $X - 1$ is the wait for the first box differing from box 1, a geometric variable with mean 2, so $\mathrm{E}(X) = 3$, matching $\frac{1}{1/4} - 1$.
- As $p \to 0$, $\mathrm{E}(X) \to \infty$: daddy penguins are rare, so you wait a long time. That matches $\frac{1}{pq} - 1$ blowing up.
- Directly, $\mathrm{P}(X = n) = p^{n-1}q + q^{n-1}p$ for $n \ge 2$. Summing $n\,\mathrm{P}(X = n)$ numerically for $p = \frac{1}{3}$ gives 3.5, and a simulation of 200 000 collectors with $p = 0.3$ gave a mean of 3.761 against the exact $\frac{1}{0.21} - 1 \approx 3.762$.

Common slips

- Using $p^{n-1} + q^{n-1}$ for $n = 1$ too, which gives a "probability" of 2 and an expectation that is 1 too large.
- Writing $\mathrm{P}(X \ge 4) = p^4 + q^4$: only the first *three* boxes need to match for a fourth box to be needed.
- Forgetting that reciprocals reverse inequalities between positive numbers in the final part.

Beyond STEP

The tail-sum formula has a continuous twin: for a non-negative random variable, $\mathrm{E}(X) = \int_0^\infty \mathrm{P}(X > x)\,\mathrm{d}x$, proved by exactly the same swap of order (Tonelli's theorem justifies it in general). The penguin problem is the two-type case of the *coupon collector's problem*; with N equally likely types the expected number of boxes is $N(1 + \frac{1}{2} + \cdots + \frac{1}{N})$,

roughly $N \ln N$. Both ideas appear in Grimmett and Welsh, *Probability: An Introduction*, and in Blitzstein and Hwang, *Introduction to Probability*.

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